PURUSHOTAM KUMAR RAUNIYAR

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Portfolio | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Software Engineer with professional experience building production web applications, REST APIs, testing workflows, and CI/CD pipelines, complemented by hands-on AI/ML project experience. Developed machine-learning systems for atmospheric event prediction and image classification using Python, PyTorch, scikit-learn, XGBoost, Pandas, and NumPy, and built LLM applications using RAG, embeddings, vector retrieval, and LLM APIs. Experienced in backend integration, model experimentation, evaluation, and Agile software development, with a B.Tech. in Computer Science and Engineering.

TECHNICAL SKILLS

AI / Machine Learning: Python, PyTorch, scikit-learn, XGBoost, LSTM, ResNet18, Machine Learning, Deep Learning, Computer Vision, Image Classification, Feature Engineering, Model Evaluation, SHAP
LLM / NLP: Large Language Models, RAG, Text Embeddings, Vector Retrieval, FAISS, Groq API, Llama 3.1 8B Instant, FastAPI
Data: Pandas, NumPy, SQL, MySQL, MongoDB
Backend & APIs: Node.js, Express.js, FastAPI, REST APIs
Software Engineering: Java, JavaScript, React.js, Git, GitHub, Docker, CI/CD, Agile/Scrum
Testing: Jest, React Testing Library, Postman, Unit Testing, API Testing

PROFESSIONAL EXPERIENCE

Software Engineer Intern — Simutopia Pvt. Ltd. *Sep 2024 – Jun 2025*
React.js, JavaScript, A-Frame, Node.js, Express.js, REST APIs, Git, GitHub, CI/CD, JSDoc

- Developed production VR web applications using React.js and A-Frame, building 10+ reusable components across multiple modules.
- Built interactive analytics and Server Health dashboards for visualizing application and backend infrastructure metrics.
- Resolved 20+ frontend and integration bugs through debugging, validation testing, and collaborative issue resolution.
- Achieved 90% unit test coverage using Jest and React Testing Library and contributed to CI/CD deployment workflows.
- Collaborated with a 5-member Agile Scrum team, contributing 100+ pull requests to the production Interia application.

AI / MACHINE LEARNING PROJECTS

AI-Based Cloudburst Prediction System

Python, XGBoost, LSTM, Logistic Regression, SHAP, ERA5, scikit-learn, Pandas

- Developed a machine-learning pipeline to predict cloudburst events approximately 12 hours before occurrence using ERA5 atmospheric data.
- Cleaned and processed atmospheric reanalysis data and engineered physics-inspired features for model training.
- Compared Logistic Regression, LSTM, and XGBoost models using Leave-One-Event-Out validation.
- Applied SHAP to analyze feature contributions and improve model interpretability.
- Evaluated model behavior across individual cloudburst events and identified XGBoost as the best-performing approach among the evaluated models.

Cloud Image Classification — Hybrid Quantum-Classical ML

Python, PyTorch, ResNet18, PennyLane, Qiskit, NumPy, Pandas, Matplotlib

- Developed image preprocessing and classification pipelines for a seven-class cloud image dataset using Python and PyTorch.
- Built and evaluated classical and hybrid quantum-classical image classification architectures.
- Compared model performance across training, validation, and test datasets to assess the effectiveness of hybrid architectures.
- Investigated quantum feature contributions through an adaptive alpha mechanism and contributed to research manuscript preparation.
- Analyzed experimental results to identify practical limitations of hybrid quantum models relative to classical architectures..

YouTube Summarizer RAG Chatbot

React/Vite, FastAPI, Python, YouTube Transcript API, Embeddings, FAISS, LLM

- Built an AI-powered YouTube summarization application with a React/Vite frontend and FastAPI backend.
- Extracted YouTube transcripts and processed source text through chunking and embedding-based retrieval.
- Integrated FAISS for vector-based retrieval of relevant transcript content.
- Connected retrieved context with an LLM generation workflow to produce summaries and responses from video content.
- Designed the application as an end-to-end AI workflow connecting frontend requests, transcript processing, retrieval, and LLM generation.

GPT-Style Conversational AI Application

React 19, Vite, Node.js, Express.js, MongoDB,

- Built a full-stack conversational AI application using React/Vite and Express.js, integrating Llama 3.1 8B Instant through the Groq API.
- Designed REST API workflows for conversational interactions and persistent conversation threads, with MongoDB/Mongoose storing messages, titles, and timestamps.
- Developed a responsive chat interface with Markdown rendering and syntax highlighting for model-generated responses..

EDUCATION

Andhra University — B.Tech., Computer Science & Engineering *Graduated 2026*

CGPA: 8.33 / 10 | ICCR Scholarship Awardee – Government of India

Relevant Coursework: Machine Learning, Artificial Intelligence, Large Language Models, Reinforcement Learning, Data Structures & Algorithms

CERTIFICATIONS AND ACHIEVEMENTS

- Accenture Learning Program — AI/ML, GenAI
- JPMorgan Chase & Co. — Software Engineering Virtual Experience, Forage
- Runner-up — Smart India Internal Hackathon 2024
- ICCR Scholarship Awardee — Government of India